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SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech / MBA Tech.

Stream/s : Data Science

Subject: MLOps

Date: 09/12/2024

Marks: 100

Year: IV

Semester: VII

Time: 3 hrs (10:00am to 1:00pm)

No. of Pages:3

Final Examination (2024-25)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right-hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-3 SO-1 BL- 1	a.	What is the difference between MLOps and DevOps?	[5]
CO-1 SO-2 BL- 2	b.	Why Should an organization have mature MLOps?	[5]
CO-2 SO-2 BL-1	c.	What is the A/B split approach of model evaluation?	[5]
CO-3 SO-2 BL-1	d.	Describe various components that constitute the MLOps Process with a neat diagram.	[5]
Q2 CO-1 SO-1 BL-2	A	Describe various components of Kubernetes architecture with a neat diagram.	[10]
Q2 CO-3 SO-3 BL-3	B	What is a CICD pipeline? Explain various stages involved in the MLOps pipeline.	[10]

Q3 CO-2 SO-2 BL-3	A	Write short notes on the following terminology related to **i.** MLFLow **ii.** Docker Container **iii.** Airflow **iv.** DVC	[10]
Q3 CO-3 SO-1 BL-2	B	Write short notes on the following terminology related to **i.** Infrastructure as a code **ii.** Kubernetes POD **iii.** Microservices **V.** Orchestration	
Q4 CO-3 SO-2 BL-1	A	Your team is starting up a new ML Project and you have been asked to set up code Repo (on Azure or GitHub) for the same. Please Perform the needed steps as below and upload a screenshot of the commands as answers. 1: Initialize the git repo on your local System. 2: Configure your user details in the git client. 3: Prepare .gitignore, readme_file for Python Project 4: Create 3 files created in the root project directory – {your_name}.py and {your sur_name}.py & {your_sap_id}.py 3: Commit {your_name}.py with the message "{your name} committing this change for term test 1" 4: Discard {your sur_name}.py from the working directory. 5: Push {your_sap_id}.py along with. gitignore, readme_file to your remote repo.	[10]
Q4 CO-2; SO-3; BL-3	B	You have been hired as an MLOps engineer in an organization and assigned the task of creating Classical machine learning operations architecture and process flow. Please sketch it and explain in details.	[10]

Q5 CO-2; SO-2; BL-2	A	Explain the significance of source code control in ML Project, its history, and its role in version control and collaboration.	[10]
Q5 CO-3; SO-2; BL-2	B	What is cloud computing? Explain the role of cloud computing in implementing the MLOps Process.	
Q6 CO-1 SO-1 BL-2	A	Explain various Personas for an ML platform in detail.	[10]
Q6 CO-2 SO-2 BL-2	B	Describe a Docker container's lifecycle with commands.	[10]
Q7 CO-3 SO-2 BL-4	A	Explain Different Model packaging and Deployment types and platforms for hosting models?.	[10]
Q7 CO-1 SO-2 BL-3	B	What are Feature Stores? Explain it's architecture in detail .	[10]